GM UNIVERSITY
(Established under the Karnataka State Act No. 19 of 2023)
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Decision Tree (6.1,6.2)

Decision Tree Learning Model

- Decision tree learning model, one of the most popular supervised predictive learning models, classifies data instances with high accuracy and consistency. The model performs an inductive inference that reaches a general conclusion from observed examples. This model is variably used for solving complex classification applications.

- Decision tree is a concept tree which summarizes the information contained in the training dataset in the form of a tree structure. Once the concept model is built, test data can be easily classified.

Structure of a Decision Tree

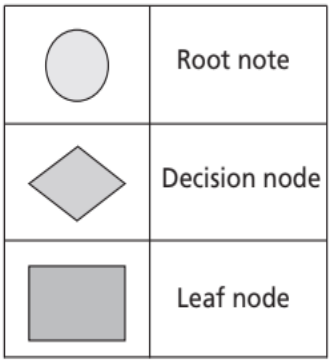


Figure 6.1: Nodes in a Decision Tree

BUILDING THE TREE

Goal: Construct a decision tree with the given training dataset. The tree is constructed in a top-down fashion. It starts from the root node. At every level of tree construction, we need to find the best split attribute or best decision node among all attributes. This process is recursive and continued until we end up in the last level of the tree or finding a leaf node which cannot be split further. The tree construction is complete when all the test conditions lead to a leaf node. The leaf node contains the target class or output of classification.

Output: Decision tree representing the complete hypothesis space.

KNOWLEDGE INFERENCE OR CLASSIFICATION

Goal Given a test instance, infer to the target class it belongs to.

Classification Inferring the target class for the test instance or object is based on inductive inference on the constructed decision tree. In order to classify an object, we need to start traversing the tree from the root. We traverse as we evaluate the test condition on every decision node with the test object attribute value and walk to the branch corresponding to the test's outcome. This process is repeated until we end up in a leaf node which contains the target class of the test object.

Output Target label of the test instance.

Advantages of Decision Trees

- Easy to model and interpret
- Simple to understand
- The input and output attributes can be discrete or continuous predictor variables.
- Can model a high degree of nonlinearity in the relationship between the target variables and the predictor variables
- Quick to train
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Some of the issues that generally arise with a decision tree learning are that:

Disadvantages of Decision Trees

- It is difficult to determine how deeply a decision tree can be grown or when to stop growing it.
- If training data has errors or missing attribute values, then the decision tree constructed may become unstable or biased.
- If the training data has continuous valued attributes, handling it is computationally complex and has to be discretized.
- A complex decision tree may also be over-fitting with the training data.
- Decision tree learning is not well suited for classifying multiple output classes.
- Learning an optimal decision tree is also known to be NP-complete.

Fundamentals of Entropy

- Entropy is the amount of uncertainty or randomness in the outcome of a random variable or an event. Moreover, entropy describes about the homogeneity of the data instances. The best feature is selected based on the entropy value. For example, when a coin is flipped, head or tail are the two outcomes, hence its entropy is lower when compared to rolling a dice which has got six outcomes.

- Higher the entropy → Higher the uncertainty
- Lower the entropy → Lower the uncertainty

$$\text{Entropy_Info}(P) = \text{Entropy_Info}(P_1 \ldots P_n)$$
$$= -(P_1 \log_2(P_1) + P_2 \log_2(P_2) + \ldots\ldots + P_n \log_2(P_n))$$

Decision Tree Induction Algorithms

- There are many decision tree algorithms, such as ID3, C4.5, CART, CHAID, QUEST, GUIDE, CRUISE, and CTREE, that are used for classification in real-time environment. The most commonly used decision tree algorithms are ID3 (Iterative Dichotomizer 3), developed by J.R Quinlan in 1986.

- The accuracy of the tree constructed depends upon the selection of the best split attribute. Different algorithms are used for building decision trees which use different measures to decide on the splitting criterion. Algorithms such as ID3, C4.5 and CART are popular algorithms used in the construction of decision trees. The algorithm ID3 uses 'Information Gain' as the splitting criterion.

ID3 Tree Construction

Algorithm 6.2: Procedure to Construct a Decision Tree using ID3

1. Compute Entropy_Info Eq. (6.8) for the whole training dataset based on the target attribute.
2. Compute Entropy_Info Eq. (6.9) and Information_Gain Eq. (6.10) for each of the attribute in the training dataset.
3. Choose the attribute for which entropy is minimum and therefore the gain is maximum as the best split attribute.
4. The best split attribute is placed as the root node.
5. The root node is branched into subtrees with each subtree as an outcome of the test condition of the root node attribute. Accordingly, the training dataset is also split into subsets.
6. Recursively apply the same operation for the subset of the training set with the remaining attributes until a leaf node is derived or no more training instances are available in the subset.

$$\text{Information_Gain}(A) = \text{Entropy_Info}(T) - \text{Entropy_Info}(T, A) \quad (6.10)$$

It can be noted that as entropy increases, information gain decreases. They are inversely proportional to each other.

Example of ID3 Construction

Example 6.3: Assess a student's performance during his course of study and predict whether a student will get a job offer or not in his final year of the course. The training dataset T consists of 10 data instances with attributes such as 'CGPA', 'Interactiveness', 'Practical Knowledge' and 'Communication Skills' as shown in Table 6.3. The target class attribute is the 'Job Offer'.

Table 6.3: Training Dataset T

S.No.	CGPA	Interactiveness	Practical Knowledge	Communication Skills	Job Offer
1.	≥9	Yes	Very good	Good	Yes
2.	≥8	No	Good	Moderate	Yes
3.	≥9	No	Average	Poor	No
4.	<8	No	Average	Good	No
5.	≥8	Yes	Good	Moderate	Yes
6.	≥9	Yes	Good	Moderate	Yes
7.	<8	Yes	Good	Poor	No
8.	≥9	No	Very good	Good	Yes
9.	≥8	Yes	Good	Good	Yes
10.	≥8	Yes	Average	Good	Yes

Step 1:
Calculate the Entropy for the target class 'Job Offer'.

$$= -\left[\frac{7}{10}\log_2\frac{7}{10} + \frac{3}{10}\log_2\frac{3}{10}\right] = -(-0.3599 + -0.5208) = 0.8807$$

Iteration 1:

Step 2:
Calculate the Entropy_Info and Gain(Information_Gain) for each of the attribute in the training dataset.

Table 6.4 shows the number of data instances classified with Job Offer as Yes or No for the attribute CGPA.

CGPA	Job Offer = Yes	Job Offer = No	Total	Entropy
≥9	3	1	4	
≥8	4	0	4	0

CGPA	Job Offer = Yes	Job Offer = No	Total	Entropy
<8	0	2	2	0

Entropy_Info(T, CGPA)

$$\text{Entropy_Info}(T, \text{CGPA}) = \frac{4}{10}\left[-\frac{3}{4}\log_2\frac{3}{4} - \frac{1}{4}\log_2\frac{1}{4}\right] + \frac{4}{10}\left[-\frac{4}{4}\log_2\frac{4}{4} - \frac{0}{4}\log_2\frac{0}{4}\right] + \frac{2}{10}\left[-\frac{0}{2}\log_2\frac{0}{2} - \frac{2}{2}\log_2\frac{2}{2}\right]$$

$$= \frac{4}{10}(0.3111 + 0.4997) + 0 + 0$$

$$= 0.3243$$

$$\text{Gain (CGPA)} = 0.8807 - 0.3243$$

$$= 0.5564$$

Table 6.5 shows the number of data instances classified with Job Offer as Yes or No for the attribute Interactiveness.

Interactiveness	Job Offer = Yes	Job Offer = No	Total	Entropy
YES	5	1	6	
NO	2	2	4	

Entropy_Info(T,Interactiveness)

$$\text{Entropy_Info}(T, \text{Interactiveness}) = \frac{6}{10}\left[-\frac{5}{6}\log_2\frac{5}{6} - \frac{1}{6}\log_2\frac{1}{6}\right] + \frac{4}{10}\left[-\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4}\right]$$

$$= \frac{6}{10}(0.2191 + 0.4306) + \frac{4}{10}(0.4997 + 0.4997)$$

$$= 0.3898 + 0.3998 = 0.7896$$

$$\text{Gain(Interactiveness)} = 0.8807 - 0.7896$$

$$= 0.0911$$

Entropy Information for Practical Knowledge

Practical Knowledge	Job Offer = Yes	Job Offer = No	Total	Entropy
Very Good	2	0	2	0
Average	1	2	3	

Practical Knowledge	Job Offer = Yes	Job Offer = No	Total	Entropy
Good	4	1	5	

Entropy_Info(T, Practical Knowledge)

$$= \frac{2}{10}\left[-\frac{2}{2}\log_2\frac{2}{2} - \frac{0}{2}\log_2\frac{0}{2}\right] + \frac{3}{10}\left[-\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3}\right] + \frac{5}{10}\left[-\frac{4}{5}\log_2\frac{4}{5} - \frac{1}{5}\log_2\frac{1}{5}\right]$$

$$= \frac{2}{10}(0) + \frac{3}{10}(0.5280 + 0.3897) + \frac{5}{10}(0.2574 + 0.4641)$$

$$= 0 + 0.2753 + 0.3608$$

$$= 0.6361$$

$$\text{Gain(Practical Knowledge)} = 0.8807 - 0.6361$$

$$= 0.2446$$

shows the number of data instances classified with Job Offer as Yes or No for the attribute Communication Skills.

Entropy Information for Communication Skills

Communication Skills	Job Offer = Yes	Job Offer = No	Total
Good	4	1	5
Moderate	3	0	3
Poor	0	2	2

Entropy_Info(T, Communication Skills)

$$\text{Entropy_Info}(T, \text{Communication Skills})$$

$$= \frac{5}{10}\left[-\frac{4}{5}\log_2\frac{4}{5} - \frac{1}{5}\log_2\frac{1}{5}\right] + \frac{3}{10}\left[-\frac{3}{3}\log_2\frac{3}{3} - \frac{0}{3}\log_2\frac{0}{3}\right] + \frac{2}{10}\left[-\frac{0}{2}\log_2\frac{0}{2} - \frac{2}{2}\log_2\frac{2}{2}\right]$$

$$= \frac{5}{10}(0.5280 + 0.3897) + \frac{3}{10}(0) + \frac{2}{10}(0)$$

$$= 0.3609$$

$$\text{Gain(Communication Skills)} = 0.8813 - 0.36096$$

$$= 0.5203$$

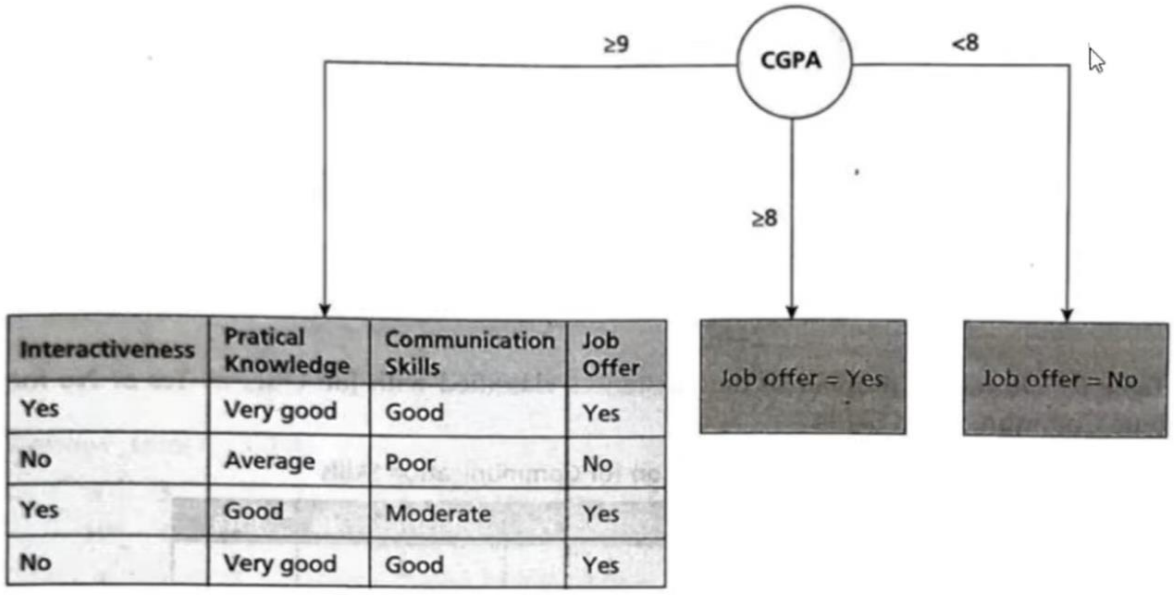
The Gain calculated for all the attributes is shown in Table 6.8:

Table 6.8: Gain

Attributes	Gain
CGPA	0.5564
Interactivityness	0.0911
Practical Knowledge	0.2246
Communication Skills	0.5203

Step 3: From Table 6.8, choose the attribute for which entropy is minimum and therefore the gain is maximum as the best split attribute.

The best split attribute is CGPA since it has the maximum gain. So, we choose CGPA as the root node. There are three distinct values for CGPA with outcomes ≥9, ≥8 and <8. The entropy value is 0 for ≥8 and <8 with all instances classified as Job Offer = Yes for ≥8 and Job Offer = No for <8. Hence, both ≥8 and <8 end up in a leaf node. The tree grows with the subset of instances with CGPA ≥9 as shown in Figure 6.3.



Interactiveness	Pratical Knowledge	Communication Skills	Job Offer
Yes	Very good	Good	Yes
No	Average	Poor	No
Yes	Good	Moderate	Yes
No	Very good	Good	Yes

Figure 6.3: Decision Tree After Iteration 1

Iteration 2:

In this iteration, the same process of computing the Entropy_Info and Gain are repeated with the subset of training set. The subset consists of 4 data instances as shown in the above Figure 6.3.

Entropy_Info(T) = Entropy_Info(3, 1) =

$$= -\left[\frac{3}{4}\log_2\frac{3}{4} + \frac{1}{4}\log_2\frac{1}{4}\right]$$
$$= -(-0.3111 + -0.4997)$$
$$= 0.8108$$

$$\text{Entropy_Info}(T, \text{Interactiveness}) = \frac{2}{4}\left[-\frac{2}{2}\log_2\frac{2}{2} - \frac{0}{2}\log_2\frac{0}{2}\right] + \frac{2}{4}\left[-\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2}\right]$$
$$= 0 + 0.4997$$

$$\text{Gain(Interactiveness)} = 0.8108 - 0.4997$$
$$= 0.3111$$

Entropy_Info(T, Practical Knowledge)

$$\text{Entropy_Info}(T, \text{Practical Knowledge})$$
$$= \frac{2}{4}\left[-\frac{2}{2}\log_2\frac{2}{2} - \frac{0}{2}\log_2\frac{0}{2}\right] + \frac{1}{4}\left[-\frac{0}{1}\log_2\frac{0}{1} - \frac{1}{1}\log_2\frac{1}{1}\right] + \frac{1}{4}\left[-\frac{1}{1}\log_2\frac{1}{1} - \frac{0}{1}\log_2\frac{0}{1}\right]$$
$$= 0$$

$$\text{Gain(Practical Knowledge)} = 0.8108$$

$$\text{Entropy_Info}(T, \text{Communication Skills})$$
$$= \frac{2}{4}\left[-\frac{2}{2}\log_2\frac{2}{2} - \frac{0}{2}\log_2\frac{0}{2}\right] + \frac{1}{4}\left[-\frac{0}{1}\log_2\frac{0}{1} - \frac{1}{1}\log_2\frac{1}{1}\right] + \frac{1}{4}\left[-\frac{1}{1}\log_2\frac{1}{1} - \frac{0}{1}\log_2\frac{0}{1}\right]$$
$$= 0$$

$$\text{Gain(Communication Skills)} = 0.8108$$

Total Gain

Attributes	Gain
Interactivities	0.3111
Practical Knowledge	0.8108
Communication Skills	0.8108

Here, both the attributes 'Practical Knowledge' and 'Communication Skills' have the same Gain. So, we can either construct the decision tree using 'Practical Knowledge' or 'Communication Skills'.

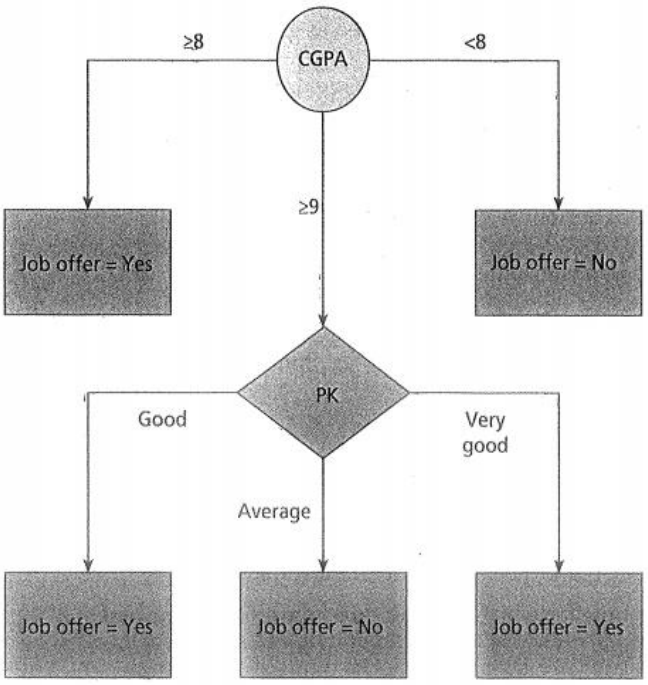


Figure 6.4: Final Decision Tree